

Senior Bioengineering student passionate about medical device engineering and the development of technologies that improve patient care and healthcare outcomes. Bringing 3 years of hands-on experience in biomedical research, cardiovascular device prototyping, experimental validation, CAD modeling, 3D printing, Python, and medical imaging analysis. Experienced in developing patient-specific anatomical models, benchtop testing systems, and data-driven engineering solutions, with a strong interest in applying technical problem-solving and rapid prototyping skills to next-generation medical device development.

EDUCATION

2026
Bachelor of Science (In Progress), Bio-Engineering, University of California, Riverside

EXPERIENCE

- NOV 2024 – PRESENT
- Undergraduate Research, University of California, Riverside, Riverside, CA
- Conducted quantitative analysis of mitral heart valve geometry across multiple patients using 3D Slicer, enabling extraction of clinically relevant anatomical parameters.
 - Processed and interpreted medical imaging datasets to identify and delineate key cardiac structures, including the mitral annulus and valve leaflets, ensuring anatomically accurate segmentation.
 - Developed patient-specific spatial models of mitral valve morphology by importing fiducial markers and applying advanced segmentation techniques (e.g., paintbrush-based region definition), supporting downstream analysis and validation of valve mechanics.

- JUN 2025 – JUL 2025
- Summer Research Program, Keck Graduate Institute, Claremont, CA
- Designed and prototyped a non-invasive LVAD balloon pump and validated performance in bench trials with measurable results.
 - Developed a SolidWorks heart test rig and produced multiple 3D printed iterations to shorten evaluation cycles.
 - Conducted experimental validation and presented technical results to professors and students to inform project direction.

- JUN 2023 – SEP 2023
- Internship, K & A Tooling, Inc., Santa Ana, CA
- Reviewed CAD drawings to confirm part marking and prevent mislabeling before shipment.
 - Conducted physical counts and inventory audits for 300 parts to ensure accurate stock records.
 - Packaged and boxed components for timely delivery, meeting shipping requirements.
 - Verified completion of production sign-offs and documentation to maintain quality compliance.

PROJECTS

- Smart Compression Sleeve for Physical Therapy
- SENIOR DESIGN CAPSTONE | SEPTEMBER 2025 (IN-PROGRESS)
- Designed and developed a wearable Smart Compression Sleeve integrating IMU, EMG, and PPG sensors to monitor motion and physiological signals for at-home physical therapy and rehabilitation.
 - Developed a real-time Flutter and Python interface to visualize live sensor data, rehabilitation feedback, and exercise performance through wireless and serial communication pipelines.
 - Created CAD models for compact sensor pods and integrated the embedded hardware system using electrical schematics, rapid prototyping, signal processing, and machine learning for motion classification and rep tracking.

- CAD Driven Designs and 3D Printing
- HOBBY | JUNE 2025 – IN PROGRESS
- Designed a CAD model and 3D printed an RC car.
 - Designed a CAD model and 3D printed figurines and fidget toys.
 - Designed electrical Block Diagrams to integrate into my CAD designs.

- Mathematical Modeling of Blood Coagulation
- PROJECTS | SEPTEMBER 2024 – JUNE 2025
- Built and validated an ODE-based MATLAB model; implemented enzyme complex kinetics (TF-VIIa, VIIIa-IXa, Va-Xa); results matched Article data and provided insights into clotting regulation.
 - Using MATLAB Simbiology programming, implemented the reaction rate using Michaelis-Menten and the first ODE to understand the metabolic reaction of ethanol.

SKILLS

Python	MATLAB	Arduino	SolidWorks
COMSOL	Microsoft Word	Microsoft Excel	Microsoft PowerPoint
NI Multisim	Raspberry pi	3D Slicer Software	3D Printing
Laser Cutting	Soldering	PCR	DNA Extraction
Gel Electrophoresis	Chromatography	Spectroscopy	Visual Studio Code